

Driver Drowsiness Detection System Using CNN

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Abstract- Consistently many individuals lose their lives because of lethal road accidents all throughout the world and drowsy driving is one of the essential mistakes of street mishaps and demise. Because of this serious problem, I set out to develop a neural network that can detect if eyes are closed, and when applied in with computer vision, to detect if a live human has had their eyes closed for more than a second. This technology is useful for safety of drivers, reduce road accidents. Neural Networks are a promising area of AI model for diminishing mishaps because of tiredness. The model fabricated was solid and by enhancing the video input, this could possibly be utilized, in actuality, applications. All things considered, the current webcam application can caution the client actually and two or three milliseconds of arriving at the shut-eye time limit. We are creating a system where we are training a dataset of approx. 30,000 images with a Convolutional Neural Network. The accuracy of our model is 96-98%.

Keywords- Convolution neural network, deep learning, drowsiness, face detection, Neural Network, road accident, safety of drivers.

I. INTRODUCTION

The National Highway Traffic Safety Administration expected 31,720 humans died in motor car visitor crashes from January through September 2021, a boom of approximately 12% from the 28,325 fatalities projected for the first nine months of 2020. The projection is the very best quantity of fatalities throughout the first nine months of any 12 months because 2006 and the best percentage growth during the primary nine months inside the Fatality Analysis Reporting System's history.

Research has discovered that going more than 20 hours without sleep is the equivalent of getting a blood-alcohol concentration of zero.08% – the U.S. Felony restrict. Given the dangers of drowsy using, the necessity of preserving drivers alert cannot be understated. This task tries to discover drowsy drivers in actual-time and then offer signals to ensure they stay awake. The existing system proposes in which the first opportunity uses a recurrent and convolutional neural community, while the second uses deep learning techniques to extract numeric

functions from snapshots, which might be brought right into a fuzzy good judgment-based totally device in a while. The minimization of the false advantageous rate received is taken into consideration as a success. %.

Although the acquired effects do no longer achieve very exceptional costs. Its accuracy while efficaciously detecting the drowsiness of the driving force, however, needs to be advanced, because the gadget alerted the driving force only in about 22% out of 61 % motion pictures wherein the subjects were drowsy.

This paper provides an answer for driver drowsiness detection with the usage of a Convolutional Neural Network. The implementation of this project makes use of a custom CNN architecture with extra statistics that has to study for clean deployment on the aspect or computationally much fewer green devices. As a result, the driver can be alerted on the proper time if the machine detects that the driver has fallen asleep earlier than something risky happens.

This technology could be useful for drivers & proprietors of commercial cars and trains, in addition to everyone interested by improved riding safety,

consisting of ordinary drivers, vehicle corporations, and car-insurance organizations.

II. RELATED WORK

The survey that's executed consists of the present technology and studies to be had associated with the topic of our project. It is a try to better understand the efforts that have long past into this area of examine, and also to apprehend in which our efforts ought to be centered on while growing this project.

This literature evaluation has been completed on the topic of the current drowsiness detection technology for full facial detection and blink detection across many techniques to perform drowsiness detection, which includes deep CNN, Computer Vision Behavioral measures, and device gaining knowledge of strategies all with unique benefits, challenges, and exceptional levels of accuracy.

One methodology is to tune down modifications in automobile behavior, consisting of the one proposed with the aid of McDonald et al [1]. He made a context-oriented and transient calculation that makes use of the guiding point, car velocities, and gas pedal positions. These values are surpassed right into a Bayesian Organization which comes to a decision in the event that a motive force indicates attributes of lazy conduct.

The calculation became found to have decrease bogus high-quality prices than PERCLOS [7] strategies, which predicts laziness dependent on eyelid traits and examples. The focal factor from this evaluation changed into that to assume appropriately, the putting of the condition is crucial. The statistics that it catches over a past 10-second duration is essential in expertise whether or not the person is in chance of tiredness associated path takeoffs.

Tayab Khan et al. [2] advise a picture-based framework for the detection and reputation of drowsiness primarily based on someone's eyes. Such a framework may be instrumental in a multitude of eventualities, which include driving force drowsiness detection, and hence has the potential to keep lives. This framework constantly video display units a driving force's condition in actual-time by using the use of a video digicam mounted in the front of the

individual. From the video, we use an image-based noninvasive technique to discover the eyes of the motive force and classify them as being open or no longer. However, the proposed method works best inside the day time; its model to night time could be explored in future paintings with greater stat-of-the-art face and eye detection algorithms.

Wanghua Deng's [5] endorse a machine known as DriCare, which detects the drivers' fatigue reputes, inclusive of yawning, blinking, and length of eye closure, the use of video snap shots, without equipping our bodies with gadgets. Owing to the shortcomings of preceding algorithms, they introduce a brand-new face-tracking algorithm to improve the monitoring accuracy.

Further, we designed a brand-new detection method for facial regions primarily based on 68 key factors. Then they use those facial areas to assess the drivers' kingdom. By combining the functions of the eyes and mouth, DriCare can alert the driver the usage of a fatigue caution. The experimental outcomes showed that DriCare completed round ninety-two% accuracy.

Shakeel et al. [2] applied Mobile Net-SSD design to put together a custom dataset of 350 photos. The model changed into fit for conducting a Mean Normal Accuracy of 0.84. The framework was cost-powerful and effective as the calculation can be conveyed in an Android machine and the digital camera circulation could be organized constantly. Celona et al. [4] proposed a dream based Perform a couple of obligations Driver Observing Structure that breaks down eyes, mouth, additionally, the posture of the head all of the even as to foresee the diploma of the sluggishness. This evaluate turned into led on the NTHU [dataset].

The System here recommends by Parikshith, Nikhil C Bhat2, Pavan Vittal [13] is a deep learning approach. A single layered CNN model is built so that it will expect the outcome. Rather than having a dependence on the chance, we rely on the dataset and the neural community wherein the weights are decided. With the cutting-edge efficiency being ninety-two% the efficiency may be expanded via growing the scale of the dataset.

III. METHODOLOGY

The proposed system uses the application of convolution neural network. A convolutional neural community is a unique kind of deep neural network which plays extraordinarily role for image class functions.

IV. PROPOSED SYSTEM

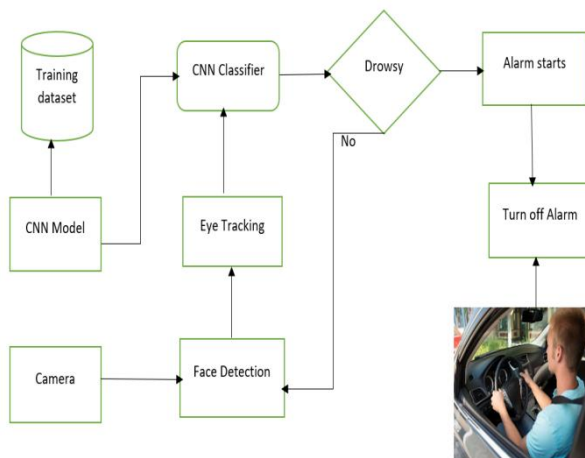


Fig 1. Proposed System.

In the above-described proposed system we are taking some 30,000 images that will be trained and then extracted by using CNN model, we add 3 convolution layers and extract features then with pooling layer we select features and then pass it to fully connected layer. After preparing model we apply that model to the webcam through which face detection and eye tracking will be performed to check whether driver is drowsy or not. If drowsiness detected then alarm starts otherwise it stops. We try to obtain better accuracy by giving more training dataset.

A CNN basically includes an input layer, an output layer, and a hidden layer that can have multiple layers. A convolution operation is achieved on these layers using a filter out that plays 2D matrix multiplication at the layer and filter out.

The CNN version architecture includes the following layers (Fig 1):

- Convolutional layer; 32 nodes, kernel size three
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- Convolutional layer; sixty-four nodes, kernel size three
- Fully connected layer; 128 nodes

The very last layer is likewise a completely related layer with 2 nodes. A Relu activation characteristic is used in all of the layers except the output layer wherein we used SoftMax.

The activation function (relu function) applied as follows-

$$\max(0, x)$$

At last layer soft-max function is used as:

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

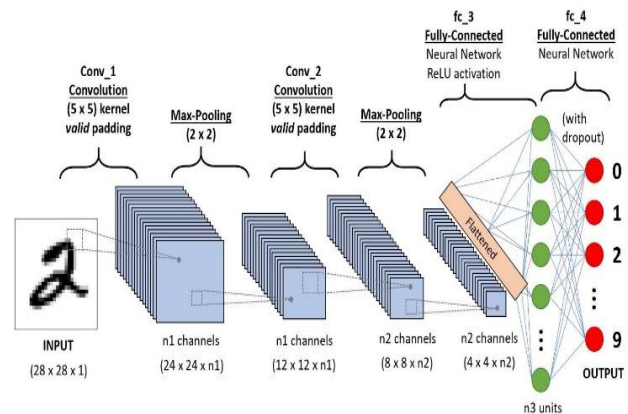


Fig 2. CNN version architecture.

IV. ANALYSIS AND INTERPRETATION

To deal with the research query and related studies troubles as defined formerly, the research technique we use is to research and evaluate a diffusion of techniques for summarizing pictures via photo type. Overall, a few strategies we protected. Then take a look at starts off evolved by using with the aid of applying convenient neural community algorithm for implementation. The aim is to create a well-known version through which other proposed methodologies may be compared.

V. SYSTEM DESIGN

The model is designed in such a way that eyes are detected and predicted that whether person is drowsy or not. If drowsiness detected then an alarm alerts and wake a person. To implement this design webcam will be used for capturing images.

It can be seen that the camera is used for tracking the driver's face constantly and upon detection of drowsiness or fatigue, the gadget within the dashboard generates a voice alert type caution to the driver.

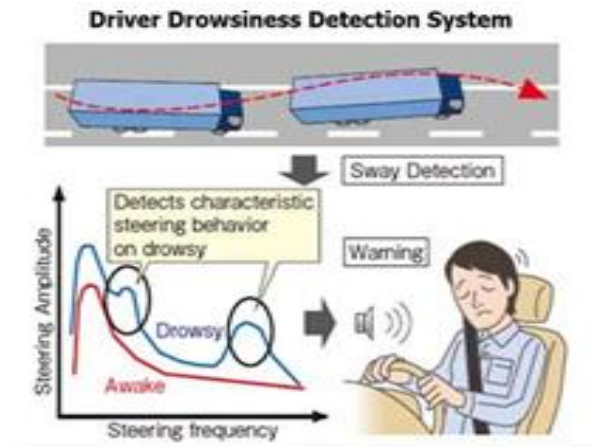


Fig 3. Tracking the driver's face constantly and upon detection of drowsiness or fatigue.

From the above-noted techniques, the technique using eye function only is the most reliable and generates extra correct outcomes than different strategies.

The present machine makes use of PERCLOS, AdaBoost and CNN to hit upon drowsiness. The device is now not advanced to send alerts to drivers. In the proposed device, we use CNN to categories the driver's pix. A digicam may be positioned in front of the motive force with the intention to take snap shots of the driver whilst driving the car.



Fig 4. shots of the driver whilst driving the car.

These images (Fig 3) will be given to the detection system where CNN will classify them into one-of-a-kind lessons- drowsy and not drowsy. A version may be built the usage of CNN and dataset containing the pictures of the driver. This dataset will include both drowsy in addition to non-drowsy photos.

Once the model is skilled, device will begin taking pictures snap shots of driving force at the same time as riding. If for a given time percentage of program language period, the share of detected drowsy photographs whilst using is greater than a threshold percent of drowsiness then the driving force is observed drowsy and alarm rings.

The process to implement this model is defined as:

- Step1-Preprocessing the dataset
- Step2-Feature Extraction
- Step 3-Feature Selection
- Step 4-Classifying model creation
- Step 5-Applying the model

V. EXPERIMENTAL RESULTS

The dataset contains approximate 30,000 images of closed and open eyes in which is divided into training and testing features.

We extract the dataset and select the features required for classifying the model. The result will be obtained gives following accuracy:

Epoch 1/5

846/846 [=====] - 135s
159ms/step - loss: 0.1170 - accuracy: 0.9584 -
val_loss: 0.0822 - val_accuracy: 0.9724

Epoch 2/5

846/846 [=====] - 112s
132ms/step - loss: 0.0644 - accuracy: 0.9769 -
val_loss: 0.0679 - val_accuracy: 0.9747

Epoch 3/5

846/846 [=====] - 121s
143ms/step - loss: 0.0503 - accuracy: 0.9820 -
val_loss: 0.0772 - val_accuracy: 0.9724

Epoch 4/5

846/846 [=====] - 108s
128ms/step - loss: 0.0406 - accuracy: 0.9848 -
val_loss: 0.0607 - val_accuracy: 0.9824

Epoch 5/5

846/846 [=====] - 121s
143ms/step - loss: 0.0330 - accuracy: 0.9878 -
val_loss: 0.0577 - Val accuracy: 0.9820

We save the model and apply that in live detection for detecting drowsiness of driver by capturing images of drivers.

In the process of applying model, we use haar cascade model also which is used to detect objects that are eyes on face when trained with lot of negative and positive images. Then we detect for both left and right eye part using prediction model and check for drowsiness. In the final result we open webcam using Open CV library and predict for drowsiness in the condition that if person closes eyes for more than 10seconds then alarm starts and alert the driver and prevent accidents.

V. CONCLUSION AND FUTURE SCOPE

Neural Networks are a promising area of system gaining knowledge of model for reducing accidents due to drowsiness. The version built turned into reliable and with the aid of optimizing the video input, this may thoroughly be utilized in actual existence packages.

Even so, the current webcam utility is capable of alert the person very successfully and within a pair millisecond of accomplishing the closed eye time threshold. Increasing convolutional layers can be effective up to a point, but the dense layers can still be the distinction maker by means of growing layers or wide variety of neurons. Driver-Factories or commercial automobile corporations should use it as a manner to overwork employees, in preference to supply them proper relaxation.

Drivers might also depend upon being awoken by using the alert and feel more secure driving at the same time as drowsy, but even an alert 1 second after falling asleep might be too past due to save you a crash. Overall, this project may be deemed a fulfillment. We created a model that may reliably be expecting whether or not a stay human's eyes are closed, and that turned into our remaining intention.

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